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ORIGINAL RESEARCH ARTICLE

Healthcare Costs and Carbon Emissions of Stage III Melanoma Surveillance Imaging

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Abstract

Objectives The aim of this study was to estimate the health system cost and carbon emissions of diagnostic imaging tests undertaken by patients on different surveillance schedules for follow-up of stage III melanoma. We also aimed to demonstrate how different monetary valuations of carbon emissions affect overall cost.

Methods We conducted a retrospective analysis of administrative data from the Melanoma Institute Australia's Melanoma Research Database for patients diagnosed with stage III melanoma between 2000 and 2014 and followed them until 2023. Imaging tests (computed tomography [CT], positron emission tomography [PET], PET–CT, ultrasound, X-ray, and magnetic resonance imaging [MRI]) undertaken during follow-up were described. Healthcare costs were estimated per patient-year using data from the Medicare Benefits Schedule. Carbon emissions from tests and transport were estimated per patient-year using life cycle assessment and valued using New South Wales carbon values.

Results Overall, 553 patients were included in this study: 115 in the 3–6-monthly surveillance imaging group, 273 in the 12-monthly surveillance imaging group, and 165 in the no routine imaging surveillance group. Healthcare costs and carbon emissions were highest in the 3–6-monthly group (Australian dollar [AUD] $1098 and 226 kg carbon dioxide equivalent emissions [CO_2-e] per patient-year) followed by the 12-monthly imaging group (AUD $767 and 150 kg CO_2-e per patient-year), and the no routine imaging group (AUD $319 and 50 kg CO_2-e per patient-year). When carbon emissions were valued in Australian dollars they accounted for 1.8–2.6% of total costs.

Conclusions More frequent surveillance imaging of patients with stage III melanoma is associated with higher healthcare costs and environmental impacts, the latter of which are responsible for a small proportion of total costs when valued in dollars.

Key Points for Decision Makers

We found that more frequent imaging schedules are associated with higher costs and carbon emissions. When carbon emissions were valued in dollars they accounted for a small proportion of the total cost.

Research on methods to include environmental impact in health economic evaluations should continue. This work should reflect the possible limitations of valuing climate change impacts in dollars and the methods that rely on this.

1 Introduction

Climate change is having a devastating impact on human health. Higher temperatures, more extreme weather events, changes to the distribution of infectious diseases, and water and food insecurity all directly threaten the health and wellbeing of people in Australia and around the world [1, 2]. Despite the Paris Agreement's goal of limiting multi-year average warming to 1.5 °C or well below 2 °C, temperatures continue to rise, with 2024 being the hottest year on record and the first to exceed 1.5 °C of warming [3]. To prevent the worst health impacts of climate change, immediate reductions in greenhouse gas emissions are needed in all sectors.

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Globally, healthcare accounts for about 4% of all emissions, and in Australia it is estimated to account for between about 5% and 7% of national emissions [4–7]. There is growing interest in reducing the health sector's carbon footprint and making healthcare more environmentally sustainable, with five principles guiding how this can be carried out: tackling causes and inequalities of disease; empowering patients to take a greater role in managing their health; streamlining care systems to avoid wasteful care; prioritising healthcare with lower environmental impact; and promoting efficient management of buildings, equipment, energy, water, and waste [8]. Related to the fourth principle—prioritising healthcare with lower environmental impact—there is interest in including environmental impacts in healthcare decision-making processes [9–12], with particular emphasis on estimating environmental impacts as part of existing health economic evaluations and frameworks [13–15].

Our case study for this exercise was on melanoma surveillance in a clinical population attending a melanoma tertiary referral centre. Australia has the highest incidence of melanoma in the world, representing 10% of all new cancers diagnosed in 2020 [16]. Stage III melanoma is characterised by metastasis to patient's lymph nodes or nearby tissues. Within stage III, the mortality risk increases as the substage increases. Patients with stage III melanoma are usually treated with lymph node dissection to remove the affected nodes, and may be offered adjuvant targeted therapy, radiation therapy, or immunotherapy on the basis of the severity of the disease, patient preference, and availability (with immunotherapy only becoming available in many countries in more recent years) [17]. After initial surgical treatment, patients have a high chance of recurrence and undergo frequent follow-up examinations (surveillance) to detect recurrences early [17]. These follow-up examinations may include the use of diagnostic imaging tests, commonly including positron emission tomography–computed tomography (PET–CT), CT tests in isolation, ultrasound, X-ray, and magnetic resonance imaging (MRI) [17]. However, the benefit of these tests to overall survival has not been demonstrated [18], and guideline recommendations on their frequency vary [19]. Australian guidelines do not specify how often PET–CT imaging should be undertaken for this patient group [17]. Because of this, frequent diagnostic imaging schedules for stage III melanoma follow-up are potentially a form of low-value care and are a compelling case study for exploring their environmental impacts in economic evaluations.

A cost-effectiveness analysis by Dieng et al. of different PET and CT imaging schedules (3–4-monthly, 6-monthly, and 12-monthly) found that 12-monthly imaging was the most cost-effective strategy for detecting and treating distant melanoma compared with no imaging [20]. This previous analysis did not consider environmental impacts.

The aim of this study was to estimate the health system cost and associated carbon emissions of diagnostic imaging tests used by patients undergoing different surveillance schedules for stage III melanoma follow-up. In addition, we aimed to demonstrate how different monetary valuations of carbon emissions affect overall cost. This work will contribute to a larger research effort to understand how environmental impacts could be included in health economic evaluations.

2 Methods

2.1 Study Design

Using a retrospective analysis of administrative data, we defined an inception cohort of patients diagnosed with stage III melanoma at the Melanoma Institute Australia (MIA) between 2000 and 2014 [20]. This cohort was the same as in the earlier study by Dieng et al. [20]. The MIA Melanoma Research Database (MRD2) was interrogated in March 2024, and data were retrieved for health service use between 1 January 2000 and 31 December 2023. The Dieng et al. study had a mean follow-up of 5 years, and in the current study we extended this to include all available data, up to a maximum of 23 years. This helps to determine costs over a lifetime time horizon and reduces the need for extrapolation. Ethics approval was granted by the Sydney Local Health District Ethics Review Committee (RPAH Zone), study 2019/ETH07272, and the Australian Institute of Health and Welfare Ethics Committee, study EO2019/1/454. No health economic analysis plan was developed for this study. In this costing study, results were reported according to the Consolidated Health Economic Evaluation Reporting Standards (CHEERS) [21] and the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) [22] statements (Appendix A and B).

2.2 Cohort

Participants in this study were the participants in the Dieng et al. study who remained eligible on the basis of updated data requested in March 2024 [20]. These were patients diagnosed with American Joint Committee on Cancer (AJCC) Seventh Edition stage III melanoma (all substages) at the MIA between 2000 and 2014, who were disease-free at their first follow-up after initial surgical treatment, and who consented to have their data included in the MRD2. AJCC Eighth Edition was not used in this study to ensure results were consistent with the Dieng et al. study. Patients eligible for the Dieng et al. study were excluded from this study if

data from March 2024 suggested they were not disease-free at their first follow-up after initial treatment. Participants with less than 1 year of follow-up data were also excluded from this study. In the Dieng et al. study, participants were grouped on the basis of the frequency of their scheduled CT, PET, or PET and CT surveillance imaging tests, into 3–4-monthly, 6-monthly, 12-monthly, and no routinely scheduled imaging. In the current study, the 3–4-monthly and 6-monthly groups were combined owing to small sample sizes. Thus, participants were grouped into three categories: 3–6-monthly, 12-monthly, and no routine scheduled imaging. Participants entered the cohort on the date of their investigative CT, PET, or PET–CT test confirming they were disease-free following initial surgical treatment (time zero). They remained in the cohort until they were diagnosed with a distant recurrence of their melanoma, were lost to follow-up or died. Participants who were lost to follow-up were censored on the date of their last known imaging event at MIA, as recorded in the MRD2. Participants were otherwise censored on 31 December 2023.

2.3 Participant Characteristics

Participant age at diagnosis, sex, postal code, melanoma substage and primary melanoma site were reported. The socioeconomic status of participants was estimated using the area level Index of Relative Socio-Economic Advantage and Disadvantage (IRSAD) from the Australian Bureau of Statistics Socio-Economic Indexes for Australia (SEIFA) 2021 dataset [23]. In these data, Australia-wide IRSAD deciles for postal areas were converted to quintiles and reported.

2.4 Surveillance Imaging Tests

We included CT, PET, PET–CT, ultrasound, X-ray and MRI surveillance tests. Imaging of multiple anatomical sites on the same day using the same modality was counted as a single test.

2.5 Healthcare Use and Costs

Healthcare costs were valued in 2024 Australian dollars. The cost of imaging tests was calculated from the perspective of the Australian health system and valued using appropriate Medicare Benefits Schedule (MBS) item numbers at 85% benefit amounts, consistent with an evaluation of outpatient care from the health system perspective. The MBS benefit for diagnostic imaging varies by anatomical site, as well as the number of images that are taken concurrently. For example, a CT of the chest and head together is covered by MBS code 57007, whereas a CT of the chest or brain only, are covered by MBS codes 56307 and 56007, respectively. MBS item numbers used in this study are reported in Appendix C.

2.6 Carbon Emissions

The carbon emissions associated with imaging tests were estimated as the sum of impacts from the test itself and from patient transport to attend the MIA. The per-minute carbon emissions of CT, ultrasound, X-ray and MRI tests in carbon dioxide equivalent emissions (CO_2-e) were obtained from an existing life cycle assessment (LCA) [24] and are reported in Appendix D. In the absence of an existing LCA, we used the carbon emissions of a CT test for both PET and PET–CT. Estimated scan durations were provided by the imaging Net Zero Clinical Lead for New South Wales Health (a practising radiographer) [25] and are reported in Appendix D. Carbon emissions per scan were estimated by multiplying per-minute emissions and estimated scan durations. Other environmental impacts, such as water use, biodiversity loss, and waste, were not included in this study.

For participants from New South Wales or the Australian Capital Territory (n = 533), we estimated carbon emissions for a return journey by car between the participant's home postal code and the MIA site, in Wollstonecraft, New South Wales, Australia. Distance was estimated using the gmapsdistance R package [26] and the Google Maps Distance Matrix API (Alphabet Inc.; Mountain View, USA). For participants from other Australian states or territories (n = 11), we estimated carbon emissions for a return flight between the relevant airport in the state's capital city and Sydney Airport (Appendix E). For participants whose postal code was not recorded (n = 9), travel-based carbon emissions were imputed using the median emissions per patient per visit (2.95 kg CO_2-e). We assumed travel was for the sole purpose of attending MIA and did not include any additional stops on the same journey.

Carbon emissions were estimated using the Australian National Life Cycle Inventory (AusLCI) (version 2.44) and ecoinvent (version 3.10) databases in SimaPro (release 9.6.0.1). Emissions were 0.37 kg CO_2-e per km for car travel and 243–273 kg CO_2-e per return flight for air travel (details are provided in Appendix E). The monetary unit cost of carbon emissions was estimated using the Central Carbon Value from the New South Wales Carbon emission in the Investment Framework Report (Australian dollar [AUD] $130 per tonne CO_2-e) [27].

2.7 Sensitivity Analyses

To demonstrate how valuations at different rates impact overall carbon costs, we reported results using low (AUD $90 per tonne CO_2-e) and high (AUD $230 per tonne CO_2-e)

values from the New South Wales Carbon emissions in the Investment Framework Report [27]. Details of these valuations are provided in Appendix F.

2.8 Data Analysis

Data were organised and analysed using R (version 4.1.4, R Foundation for Statistical Computing: Vienna, Austria) and R Studio (version 2024.09.0+375, posit: Boston, USA). We reported descriptive statistics for participant characteristics by the three imaging frequency groups. Group differences were investigated using Pearson's chi-squared test or Kruskal–Wallis rank-sum test, with appropriate *p*-values reported. The crude Poisson rate (count/person-time) of imaging tests, health system costs, and carbon emissions was calculated using the epiR package [28]. Tests were reported per 10 patient-years, and health system costs and climate change impacts were reported per patient-year. In addition, 95% confidence intervals (CIs) were reported using the Exact method.

3 Results

Of the 823 patients potentially eligible for this study, ten were excluded because updated data suggested they were not disease-free at their first follow-up appointment. An additional 260 were excluded for having less than 1 year of follow-up data. The remaining 553 patients included in this study were grouped into: 3–6-monthly imaging (n = 115; 21%), 12-monthly imaging (n = 273; 49%) and no routinely scheduled imaging (n = 165; 30%). An overview of participant selection is provided in Fig. 1.

There was very strong evidence of differences across the three patient groups in a number of important factors, including: mean age at time of the stage III melanoma diagnosis, proportions with higher AJCC Seventh Edition substages, proportions with high-risk anatomical sites for the primary melanoma, the group's overall risk of distant recurrence or death, median follow-up time and censor reason ($p < 0.001$). Details are presented in Table 1.

Patients had 5210 diagnostic imaging tests over the study period. The number and rate of these are presented in Table 2. Patients on a frequent imaging schedule had a higher number of total tests, including scheduled PET–CT and other tests. Participants in the 3–6-monthly imaging group had 28 tests per 10 patient-years (95% CI 27.0–30.0), participants in the 12-monthly imaging group had 20 tests per 10 patient-years (95% CI 19.5–20.9) and participants in the no routine imaging group had 6 tests per 10 patient-years (95% CI 5.7–6.9). The most common tests were CT and ultrasound in all three schedules.

Participants in the 3–6-monthly imaging group had the highest health system costs at AUD $1098 per patient-year (95% CI 1095.4–1101.1), followed by the 12-monthly imaging group at AUD $767 per patient-year (95% CI 765.6–768.3) and the no routine imaging group at AUD $319 per patient-year (95% CI 317.6–320.2). CT, PET, and/or PET–CT accounted for the largest proportion of health system costs, while X-ray accounted for the smallest proportion. Details are presented in Table 3.

Carbon emissions were highest in the 3–6-monthly imaging group at 226 kg CO_2-e per patient-year (95% CI 225.2–227.8), followed by the 12-monthly imaging group at 150 kg CO_2-e per patient-year (95% CI 149.8–151.0) and the no routine imaging group at 50 kg CO_2-e per patient-year (95% CI 49.1–50.1). Patient transport to and from the health facility accounted for the largest proportion of carbon emissions in all three groups and totalled about 70% of greenhouse emissions in the 3–6-monthly and 12-monthly groups and 51% in the no routine imaging group. The impact of the test itself accounted for the remaining emissions. Details are presented in Table 3.

When valued in dollars using the central CO_2-e unit cost, carbon emissions cost AUD $29 per patient-year (95% CI 29.0–29.9) in the 3–6-monthly imaging group, AUD $20 per patient-year (95% CI 19.3–19.8) in the 12-monthly imaging group and AUD $6 per patient-year (95% CI 6.3–6.6) in the no routine imaging group. As a proportion of total cost per patient-year (health system cost plus climate change impact), carbon emissions accounted for 1.8–2.6% using the central CO_2-e unit cost.

3.1 Sensitivity Analyses

The cost of carbon emissions varied depending on the unit cost applied to a kg of CO_2-e. When valued in dollars using the high CO_2-e unit cost, carbon emissions cost AUD $52 per patient-year (95% CI 51.5–52.7) in the 3–6-monthly imaging group, AUD $35 per patient-year (95% CI 34.3–34.9) in the 12-monthly imaging group and AUD $11 per patient-year (95% CI 11.2–11.6) in the no routine imaging group. Carbon emissions accounted for 3.3–4.5% of total costs using the high CO_2-e unit cost. When valued using the low CO_2-e unit cost, carbon emissions cost AUD $20 per patient-year (95% CI 20.0–20.8) in the 3–6-monthly imaging group, AUD $14 per patient-year (95% CI 13.4–13.7) in the 12-monthly imaging group, and AUD $4 per patient-year (95% CI 4.3–4.6) in the no routine imaging group. Carbon emissions accounted for 1.2–1.8% of total cost using the low CO_2-e unit cost. Details are presented in Table 3 and Fig. 2.

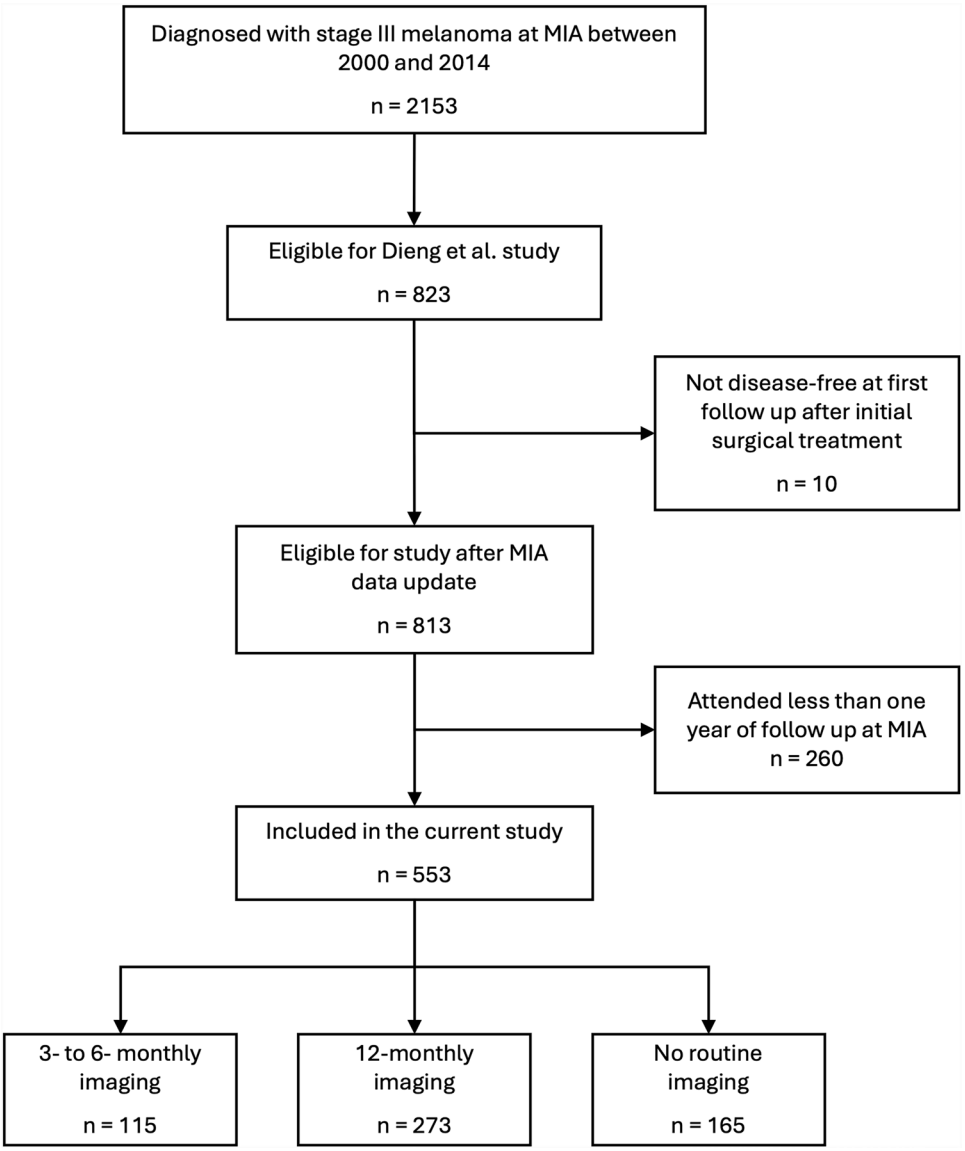


Fig. 1 Patient selection diagram

4 Discussion

This is the first Australian study to incorporate the monetary cost of carbon emissions into a cost comparison of alternative surveillance strategies. Using the example of imaging for people with stage III melanoma, we found the most frequent imaging surveillance schedule (3–6-monthly) was associated with higher health system costs and carbon emissions compared with annual imaging or no routinely scheduled imaging, as expected. Although the three groups differed in their demographic and clinical characteristics, these data provide useful information about the costs and carbon emissions associated with different frequencies of routinely scheduled cancer surveillance.

The previous cost-effectiveness analysis of this cohort found that more frequent imaging was associated with higher costs [20], consistent with our findings, and here we show that there are also higher carbon emissions with more frequent imaging. Given that there is no evidence that more frequent PET–CT imaging improves survival, these results may inform clinician and patient decisions about the frequency of imaging [18]. Although this study focused only on surveillance imaging for stage III melanoma, its findings may also inform surveillance imaging used for other types of cancer [19].

Several studies have demonstrated ways to include environmental impacts in cost analyses of healthcare interventions [9], but few have done so in the Australian context [13, 29–32]. Australian studies have reported carbon emissions alongside financial costs within life cycle assessment [29, 30], health economic [13, 32] or triple bottom line frameworks [31], and only one of these valued emissions in monetary units [32].

Table 1 Descriptive statistics of the study cohort

Characteristic	Imaging schedule			
	3–6-monthly, *most frequent* (*n* = 115)	12-Monthly, *intermediate* (*n* = 273)	No routine, *least frequent* (*n* = 165)	*P*-value[a]
Sex (%)				> 0.9
Male	73 (63)	172 (63)	107 (65)	
Female	42 (37)	101 (37)	58 (35)	
Age (years) at diagnosis of stage III melanoma, median (Q1, Q3)	57 (47, 66)	52 (42, 62)	63 (49, 74)	< 0.001[b]
Index of Relative Socio-economic Advantage and Disadvantage quintile (%)				0.2
1	11 (9.6)	26 (9.6)	18 (11)	
2	20 (18)	44 (16)	37 (23)	
3	16 (14)	40 (15)	27 (17)	
4	13 (11)	53 (19)	28 (18)	
5	54 (47)	109 (40)	48 (30)	
Missing	1	1	7	
AJCC 7th Edition melanoma sub-stage on entry into cohort (%)				< 0.001
IIIA	26 (23)	114 (42)	26 (16)	
IIIB	33 (29)	86 (32)	43 (26)	
IIIC	26 (23)	40 (15)	54 (33)	
Missing[c]	30 (26)	33 (12)	42 (25)	
Site of primary melanoma (%)				< 0.001
Head and neck	18 (16)	39 (14)	40 (24)	
Trunk	30 (26)	97 (36)	37 (23)	
Upper limbs	8 (7.0)	32 (12)	19 (12)	
Lower limbs	29 (25)	73 (27)	26 (16)	
Occult	30 (26)	32 (12)	42 (26)	
Unknown	0	0	1	
Year of entry into cohort (%)				0.13
Before or in 2010	65 (57)	183 (67)	109 (66)	
After 2010	50 (43)	90 (33)	56 (34)	
Died during study period[d] (%)				< 0.001
Yes	66 (57)	75 (27)	104 (63)	
No	49 (43)	198 (73)	61 (37)	
Distant recurrence during study period (%)				< 0.001
Yes	75 (65)	97 (36)	34 (21)	
No	40 (35)	176 (64)	131 (79)	
Censor reason (%)				< 0.001
Dead	8 (7.0)	13 (4.8)	72 (44)	
Distant recurrence	75 (65)	97 (36)	34 (21)	
Loss to follow-up[e]	32 (28)	162 (59)	58 (35)	
End of study	0	1 (0)	1(0)	
Time until censor event, years				< 0.001[b]
Median (Q1, Q3)	2.9 (1.5, 5.8)	5.5 (2.9, 8.3)	3.2 (1.8, 5.9)	
Sum	509.7	1634.3	726.1	

AJCC, American Joint Committee on Cancer; SD, standard deviation; Q1, first quartile; Q3, third quartile

[a]Pearson's Chi-squared test unless otherwise stated

[b]Kruskal–Wallis rank–sum test

[c]Confirmed as stage III but sub-stage unknown

[d]The number of participants who died at any time between their entry into the cohort and 31 December 2023, including those that were censored on the date of their distant recurrence

[e]Participants lost to follow-up includes patients discharged from follow-up at MIA or who chose to stop attending. They were censored at their last known appointment at which they were alive without distant recurrence

Table 2 Details of diagnostic imaging tests investigated in this study

	Imaging schedule					
	3–6-Monthly, *most frequent* (*n* = 115)		12-Monthly, *intermediate* (*n* = 273)		No routine, *least frequent* (*n* = 165)	
	N	*n* per 10 patient-years (95% CI)	*N*	*n* per 10 patient-years (95% CI)	*N*	*n* per 10 patient-years (95% CI)
CT	602	12 (10.9–12.8)	1490	9.1 (8.7–9.6)	173	2.4 (2.0–2.8)
PET	84	1.6 (1.3–2.0)	131	0.80 (0.67–0.95)	30	0.41 (0.28–0.59)
PET and CT	87	1.7 (1.4–2.1)	212	1.3 (1.1–1.5)	84	1.2 (0.9–1.4)
Ultrasound	235	4.6 (4.0–5.2)	1239	7.6 (7.2–8.0)	74	1.0 (0.80–1.3)
X-ray	264	5.2 (4.6–5.8)	143	0.88 (0.74–1.0)	27	0.37 (0.25–0.54)
MRI	178	3.5 (3.0–4.0)	90	0.55 (0.44–0.68)	67	0.92 (0.72–1.2)
Total	1450	28 (27.0–30.0)	3305	20 (19.5–20.9)	455	6.3 (5.7–6.9)

CI, confidence interval; CT, computed tomography; PET, positron emission tomography; MRI, magnetic resonance imaging

Table 3 Cost and environmental impact of diagnostic imaging

	Imaging schedule					
	3–6-Monthly *Most frequent* (*n* = 115)		12-Monthly *Intermediate* (*n* = 273)		No routine *least frequent* (*n* = 165)	
	Cost	Cost per patient-year (95% CI)	Cost	Cost per patient-year (95% CI)	Cost	Cost per patient-year (95% CI)
Health system costs, AUD						
CT	$297,920	$585 (582.5–586.7)	$756,554	$463 (461.9–464.0)	$88,040	$121 (120.4–122.0)
PET	$76,078	$149 (148.2–150.4)	$118,660	$73 (72.2–73.0)	$27,174	$37 (37.0–37.9)
PET and CT	$86,200	$169 (168.0–170.3)	$210,050	$129 (127.0–129.1)	$83,227	$115 (113.8–15.)
Ultrasound	$24,355	$48 (47.2–48.4)	$130,201	$80 (79.2–80.1)	$8068	$11 (10.9–11.4)
X-Ray	$11,435	$22 (22.0–22.9)	$6,321	$4 (3.8–4.0)	$1176	$2 (1.5–1.7)
MRI	$63,732	$125 (124.1–126.0)	$31,572	$19 (19.1–19.5)	$23,896	$33 (32.5–33.3)
Total	$559,728	$1098 (1095.4–1101.1)	$1,253,358	$767 (765.6–768.3)	$231,581	$319 (317.6–320.2)
Climate change impacts, kg CO_2-e						
Imaging test[a]	34,975	69 (67.9–69.4)	72,236	44 (43.9–44.5)	17,706	24 (24.0–24.7)
Transport[b]	80,456	158 (156.8–159.0)	173,592	106 (105.7–106.7)	18,317	25 (24.9–25.6)
Total	115,431	226 (225.2–227.8)	245,828	150 (149.8–151.0)	36,023	50 (49.1–50.1)
Climate change impacts, AUD[c]						
High CO_2-e valuation	$26,549	$52 (51.5–52.7)	$56,540	$35 (34.3–34.9)	$8285	$11 (11.2–11.6)
Central CO_2-e valuation	$15,006	$29 (29.0–29.9)	$31,958	$20 (19.3–19.8)	$4683	$6 (6.3–6.6)
Low CO_2-e valuation	$10,389	$20 (20.0–20.8)	$22,124	$14 (13.4–13.7)	$3242	$4 (4.3–4.6)

Costs are reported to zero decimal places, and confidence internals are reported to one decimal place

AUD, Australian dollars; CI, confidence interval; CO_2-e, carbon dioxide equivalent; SD, standard deviation

[a]Attributional climate change impact as reported by McAlister et al

[b]Impact of a return journey by car or a return flight between the relevant capital city and Sydney

[c]High valuation ($230 per tonne CO_2-e); central valuation ($130 per tonne CO_2-e); low valuation ($90 per tonne CO_2-e)

This Australian study was a cost-minimisation analysis of single-use and reusable hospital gowns which found that monetised emissions account for less than 3% of total cost [32]. Emissions in this study were valued at AUD $63 per tonne CO_2-e. A recent micro-costing study from France exploring robot-assisted total knee arthroplasty found that greenhouse gas emissions accounted for 3% of the mean total cost per patient [14]. Emissions in this study were valued at 250 € per tonne CO_2-e. An earlier study of global asthma inhaler use found that the social cost of carbon accounted for up to 39% of the total cost [15]. Emissions in this study were valued at GBP £71 per tonne CO_2-e. While comparisons

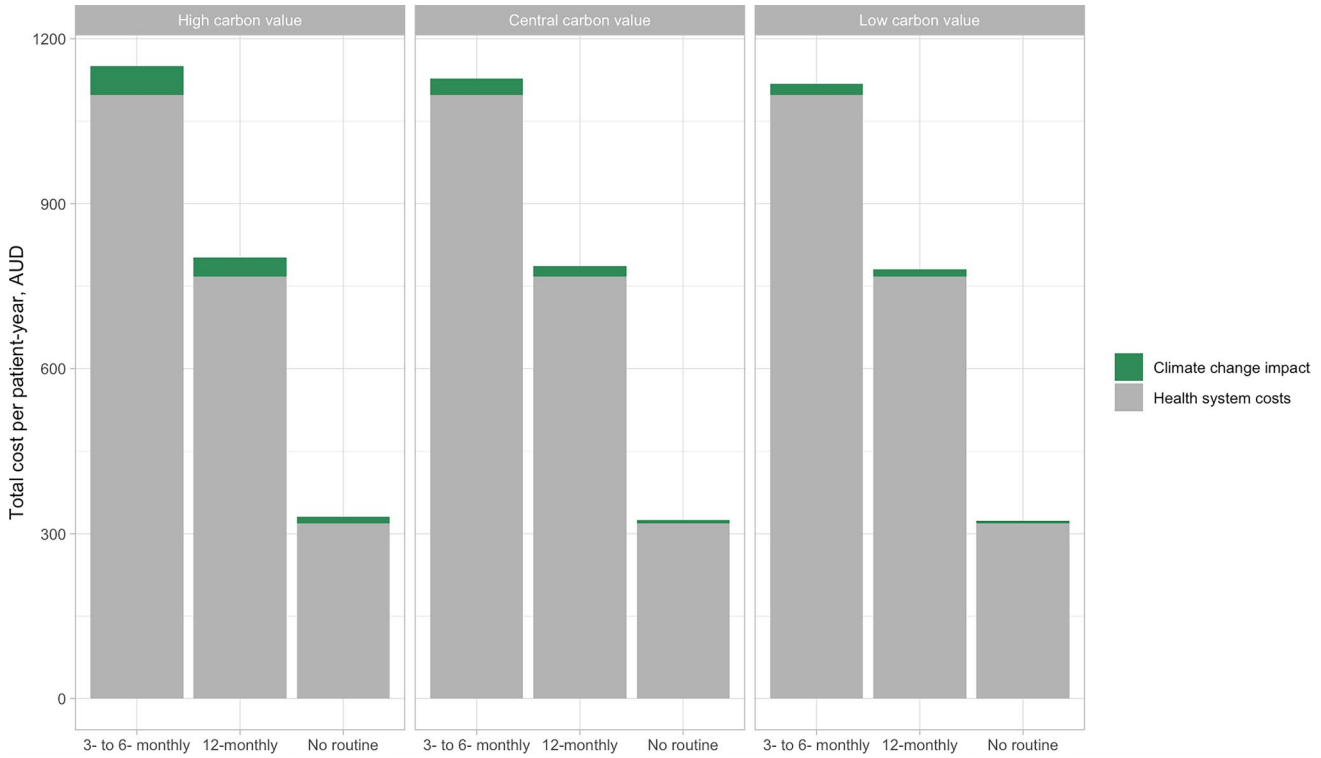


Fig. 2 The total cost of diagnostic imaging per patient-year under different carbon emission valuations

across these studies are difficult because of differences in methodology, particularly in the valuation of greenhouse gas emissions, the proportion of total cost attributable to carbon emissions is highly dependent on the valuation used, and is likely to be more influential for interventions with large environmental impacts, such as those associated with the use of metered-dose inhalers [15, 33]. The example presented in this study is likely to be of particular interest in the Australian context, using carbon valuations published by the New South Whales (NSW) Government.

Much of the existing research in this field has been carried out as part of broader work to include environmental impacts in economic evaluations, with notable examples available for evaluations in the UK [34–36]. These studies similarly found that while environmental impacts can feasibly be included within or alongside frameworks for economic evaluations, valuing impacts in monetary units is unlikely to result in substantial changes to the total or incremental cost of an intervention or model of care.

A key issue with the monetary valuation of emissions is the methodology and uncertainty underpinning the valuation used [37]. A recent study found the social cost of carbon used by the US government was 3.6 times lower than the author's preferred mean estimate of US $186 per tonne CO_2-e [38]. Another study found that estimates of the social cost of carbon in literature were too low owing to high discounting rates and model structures, and a higher amount of US $283 per tonne CO_2-e should be used [39]. Both recommended estimates are higher than the unit costs used in this study. It is possible, and even likely, that the NSW Government's valuation of carbon emissions is also too low to account for the full impact of climate change on society, and this should be investigated by future studies.

Despite our and others' findings that converting environmental impacts to monetary values has only a small impact on total costs of alternative healthcare technologies, health economic studies remain a practical framework for assessing environmental impacts. Ongoing work is exploring the best approaches to include environmental impacts within economic evaluations and health technology assessments [9–12, 40, 41]. Economic studies identify and count resource use, making them a logical place to consider environmental impacts. Doing so also ensures that environmental impacts are considered alongside health and financial outcomes and not in isolation.

While this field is still developing, a recent study by the National Institute for Health and Care Excellence (NICE) in the UK found that the public is willing for environmental impacts to be considered in evaluations when clinical and cost outcomes are equivalent [42]. Trade-offs between environmental impacts and costs may be less acceptable, and trade-offs between health outcomes and environmental impacts are unacceptable, particularly when the interventions are for a severe disease or a disease with

no other treatment options [42]. It is therefore useful for environmental impacts to be included in economic studies where these trade-offs can be described and managed appropriately.

Imaging schedules may be associated with different health outcomes and downstream healthcare use, and this should be considered in future studies. Health outcomes were not included in the current study, and caution should be taken in the interpretation of trade-offs between the costs reported and any potential health outcomes. It is unlikely that any potential health benefits would be traded off for surveillance strategies with lower carbon emissions.

Future research should explore the range of methods available to include environmental impacts in economic evaluations, including alternatives to monetary valuations of greenhouse gas emissions. This should consider the possibility that economic evaluations are not the right place to evaluate carbon emissions. We intend to use the data presented here to update an existing cost-effectiveness analysis to demonstrate some of these methods [20]. This is particularly relevant in Australia, where the recent National Health and Climate Strategy and Health Technology Assessment Policy and Methods Review recommended exploring ways to include environmental impacts in health technology assessments [6, 43].

The main strength of this study is the use of a large and detailed real-world database of health resource use and clinical data from the Melanoma Institute Australia (MIA). In addition, estimates of the climate change impact of surveillance imaging tests were obtained from a published high-quality process-based life cycle assessment and were appropriate for the Australian context. Estimation of the greenhouse gas emissions from transport were similarly robust.

The main limitation of this study was the limited scope of resources included. While data were available in the MIA database regarding other healthcare activities, such as doctor's appointments and surgeries, we excluded these owing to lack of reliable estimates of carbon emissions for these activities. As such, this study is limited to the cost and environmental impact of the imaging schedule only and does not include downstream healthcare use. Future work should consider a broader scope of impacts as more life cycle assessments of healthcare products and services are generated.

The life cycle assessment used in this analysis did not include carbon emissions from capital infrastructure, noting that emissions from these are 'amortised over long periods of time and are typically small' [24]. While this may be true, the exclusion of capital infrastructure in this study likely results in a slight underestimation of the environmental impact of imaging. Including these emissions, even if relatively minor, may produce a higher result.

This study focused on carbon emissions, recognising them as a key component of healthcare's environmental impact. Other important impacts such as biodiversity loss, water use, land use and waste were not included. While these could feasibly be reported within a cost analysis, we chose to focus only on carbon emissions owing to the urgent need to reduce greenhouse gas emissions and prevent the worst health impacts of climate change [1, 10]. This provides a policy-relevant starting point for developing economic methods to evaluate the environmental impact of healthcare. However, future work should explore how other non-greenhouse gas impacts can be meaningfully included in economic evaluations.

The study is also limited to a highly selective patient group that may not reflect the Australian population of patients with stage III melanoma undergoing surveillance screening. The results presented may therefore not be applicable to this population. In addition, as discussed, the conversions of environmental impacts into dollars are associated the significant uncertainty and may not reflect the true cost of the impacts of climate change.

5 Conclusions

More frequent surveillance imaging of patients with stage III melanoma is associated with higher healthcare costs and environmental impacts, the latter of which are responsible for a small proportion of total costs when valued in dollars. Future work should consider costs and carbon emissions alongside possible health benefits of more frequent imaging in a full economic evaluation.

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Declarations

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Data availability statement The datasets generated and analysed during the current study are available from the corresponding author on reasonable request.

Code availability The R code used for this analysis is available from the corresponding author on reasonable request.

Ethics approval Ethics approval was granted by the Sydney Local Health District Ethics Review Committee (RPAH Zone), study 2019/ETH07272, and the Australian Institute of Health and Welfare Ethics Committee, study EO2019/1/454.

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